The maths behind Einstein's equation is formidable, but the basic idea is simple, and was pithily summarized by John Wheeler: matter tells spacetime how to curve, and spacetime tells matter how to move. The left-hand side measures the curvature of spacetime, while the right-hand side characterizes energy-like quantities, including momentum, pressure, and mass.

General relativity is classical. The geometry of spacetime is unique, evolves deterministically, and can in principle be measured to arbitrary precision without disturbing it. Once quantum mechanics came along, it was perfectly natural to try to "quantize" general relativity, obtaining a quantum theory of gravity. Easier said than done. What makes relativity special is that it's a theory of spacetime rather than a theory of stuff within spacetime. Other quantum theories describe wave functions that assign probabilities to observing things at definite, well-defined locations in space and moments in time. Quantum gravity, by contrast, will have to be a quantum theory of spacetime itself. That raises some issues.

Einstein, naturally, was one of the first to appreciate the problem. In 1936, he mused on the difficulty of even imagining how to apply the principles of quantum mechanics to the nature of spacetime:

> Perhaps the success of the Heisenberg method points to a purely algebraical method of description of nature, that is to the elimination of continuous functions from physics. Then, however, we must also give up, by principle, the space-time continuum. It is not unimaginable that human ingenuity will some day find methods which will make it possible to proceed along such a path. At the present time, however, such a program looks like an attempt to breathe in empty space.

Here Einstein is contemplating Heisenberg's approach to quantum theory, which you'll remember provided a description in terms of explicit quantum jumps without trying to fill in the details about microscopic processes happening along the way. Similar worries persist if we switch to a more Schrödingerian point of view with wave functions. Presumably we would need a wave function that assigns amplitudes to different possible geometries of spacetime. But if we imagine, for example, two branches of such a wave function that describe different spacetime geometries, there is no unique way of specifying that two events in the two branches correspond to the "same" point in space-time. There is no unique map, in other words, between two different geometries.